Exp. No: 5

CLUSTERING WITH K-MEANS AND DIMENSIONALITY REDUCTION WITH PCA

Code:

```
# 1. Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import make_blobs
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score


# 2. K-Means Clustering

# 2.1 Generate Dataset
X, y = make_blobs(
  n_samples=300,
  centers=3,
  cluster_std=0.60,
  random_state=42
)
df_kmeans = pd.DataFrame(
  X,
  columns=["Feature_1", "Feature_2"]
)

print("Original K-Means Dataset Head:")
print(df_kmeans.head())
```

```
# 2.2 Elbow Method
wcss = []

for i in range(1, 11):
  kmeans = KMeans(
    n_clusters=i,
    init="k-means++",
    max_iter=300,
    n_init=10,
    random_state=42
  )

  kmeans.fit(X)
  wcss.append(kmeans.inertia_)


# 2.3 Plot Elbow Curve
plt.figure(figsize=(10, 6))

plt.plot(
  range(1, 11),
  wcss,
  marker="o",
  linestyle="--"
)
plt.title("Elbow Method for Optimal K")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.grid(True)
plt.show()


# 2.4 Apply K-Means
optimal_k = 3
```

```
kmeans = KMeans(
 n_clusters=optimal_k,
 init="k-means++",
 max_iter=300,
 n_init=10,
 random_state=42
)
clusters = kmeans.fit_predict(X)

df_kmeans["Cluster"] = clusters

print("\nK-Means Clustered Dataset:")
print(df_kmeans.head())



# 2.5 Visualize K-Means Clusters
plt.figure(figsize=(10, 8))

sns.scatterplot(
  x="Feature_1",
  y="Feature_2",
  hue="Cluster",
  data=df_kmeans,
  palette="viridis",
  s=100,
  alpha=0.8
)

plt.scatter(
  kmeans.cluster_centers_[:, 0],
  kmeans.cluster_centers_[:, 1],
  s=300,
  c="red",
  marker="X",
  label="Centroids"
)
```

```
plt.title("K-Means Clustering")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.legend()
plt.grid(True)
plt.show()


# 2.6 Calculate Silhouette Score
silhouette_avg = silhouette_score(
  X,
  clusters
)

print(
  f"\nSilhouette Score: "
  f"{silhouette_avg:.3f}"
)


# 3. Principal Component Analysis (PCA)

# 3.1 Generate 4D Dataset
X_pca, y_pca = make_blobs(
  n_samples=500,
  n_features=4,
  centers=4,
  cluster_std=1.0,
  random_state=25
)
```

```
df_pca = pd.DataFrame(
 X_pca,
 columns=[
 "Feature_1",
 "Feature_2",
 "Feature_3",
 "Feature_4"
 ]
)

df_pca["True_Cluster"] = y_pca

print("\nOriginal PCA Dataset Head:")
print(df_pca.head())

print(
  "\nOriginal PCA Dataset Shape:",
  df_pca.shape
)



# 3.2 Standardize Dataset
scaler = StandardScaler()

X_pca_scaled = scaler.fit_transform(X_pca)



# 3.3 Apply PCA
pca = PCA(n_components=2)

principal_components = pca.fit_transform(
  X_pca_scaled
)
```

```
# 3.4 Create PCA DataFrame
df_principal_components = pd.DataFrame(
  principal_components,
  columns=[
    "Principal_Component_1",
    "Principal_Component_2"
  ]
)

df_principal_components["True_Cluster"] = y_pca

print("\nPrincipal Components Head:")
print(df_principal_components.head())



# 3.5 Explained Variance
explained_variance = pca.explained_variance_ratio_

print(
  "\nExplained Variance Ratio:",
  explained_variance
)

print(
  "Total Explained Variance:",
  f"{explained_variance.sum():.3f}"
)
# 3.6 Visualize PCA Result
plt.figure(figsize=(10, 8))

sns.scatterplot(
  x="Principal_Component_1",
  y="Principal_Component_2",
  hue="True_Cluster",
```

```
data=df_principal_components,
 palette="Paired",
 s=100,
 alpha=0.8
)

plt.title("PCA - Dimensionality Reduction to 2 Components")

plt.xlabel(
  f"PC1 ({explained_variance[0] * 100:.2f}%)"
)

plt.ylabel(
  f"PC2 ({explained_variance[1] * 100:.2f}%)"
)

plt.grid(True)
plt.show()



# 4. K-Means on PCA-Reduced Data

# 4.1 Apply K-Means on PCA Data
kmeans_pca = KMeans(
  n_clusters=4,
  init="k-means++",
  max_iter=300,
  n_init=10,
  random_state=42
)

clusters_pca = kmeans_pca.fit_predict(
  principal_components
)
```

```
df_principal_components[
 "KMeans_Cluster_on_PCA"
] = clusters_pca

# 4.2 Visualize K-Means on PCA Data
plt.figure(figsize=(10, 8))

sns.scatterplot(
  x="Principal_Component_1",
  y="Principal_Component_2",
  hue="KMeans_Cluster_on_PCA",
  data=df_principal_components,
  palette="viridis",
  s=100,
  alpha=0.8
)

plt.scatter(
  kmeans_pca.cluster_centers_[:, 0],
  kmeans_pca.cluster_centers_[:, 1],
  s=300,
  c="red",
  marker="X",
  label="Centroids"
)

plt.title("K-Means Clustering on PCA-Reduced Data")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend()
plt.grid(True)
plt.show()
```

```
# 4.3 Calculate Silhouette Score
silhouette_avg_pca = silhouette_score(
  principal_components,
  clusters_pca
)

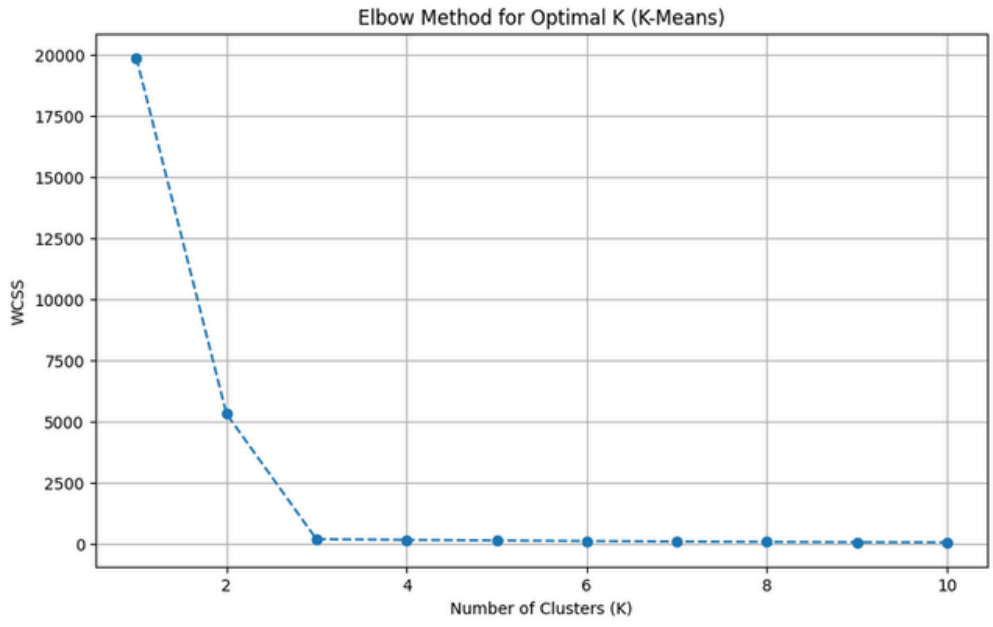
print(
  "\nSilhouette Score for K-Means on PCA-Reduced Data:",
  f"{silhouette_avg_pca:.3f}"
)


# 5. Final Output
print("\nExperiment completed successfully!")
```

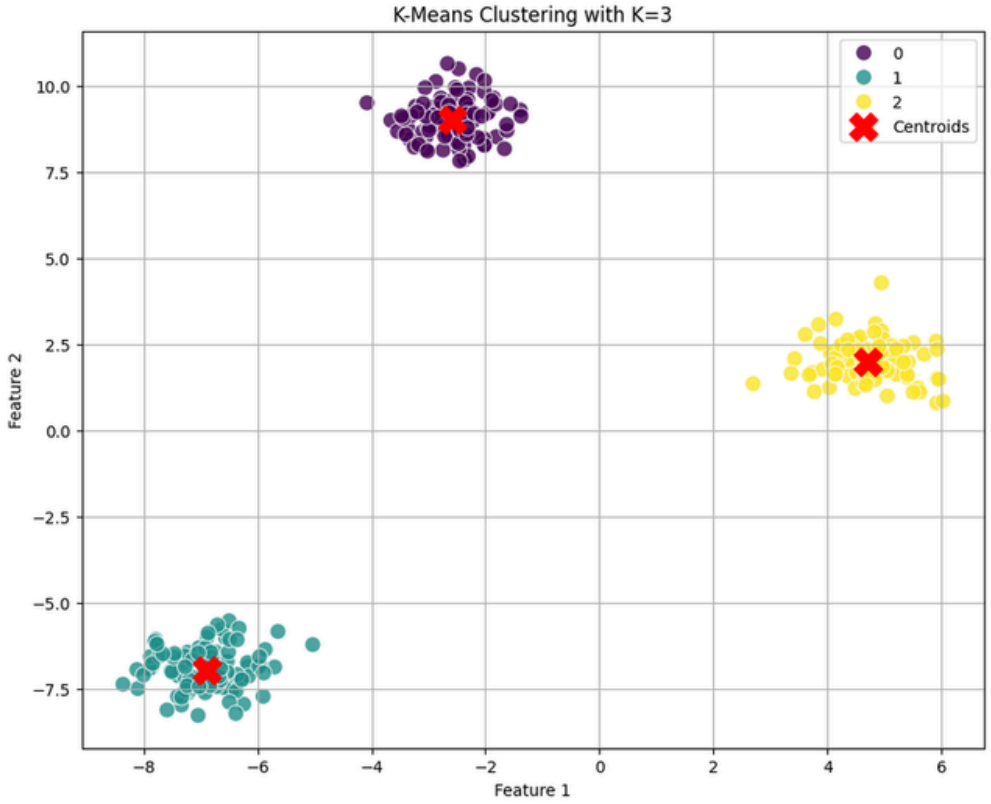
Output:

```
--- Part 1: K-Means Clustering ---

Original K-Means Dataset Head:
   Feature_1  Feature_2
0  -7.155244  -7.390016
1  -7.395875  -7.110843
2  -2.015671   8.281780
3   4.509270   2.632436
4  -8.102502  -7.484961
```



```
K-Means Clustered Dataset:
   Feature_1  Feature_2  Cluster
0  -7.155244  -7.390016        1
1  -7.395875  -7.110843        1
2  -2.015671   8.281780        0
3   4.509270   2.632436        2
4  -8.102502  -7.484961        1
```




```
Silhouette Score for K-Means (K=3): 0.908

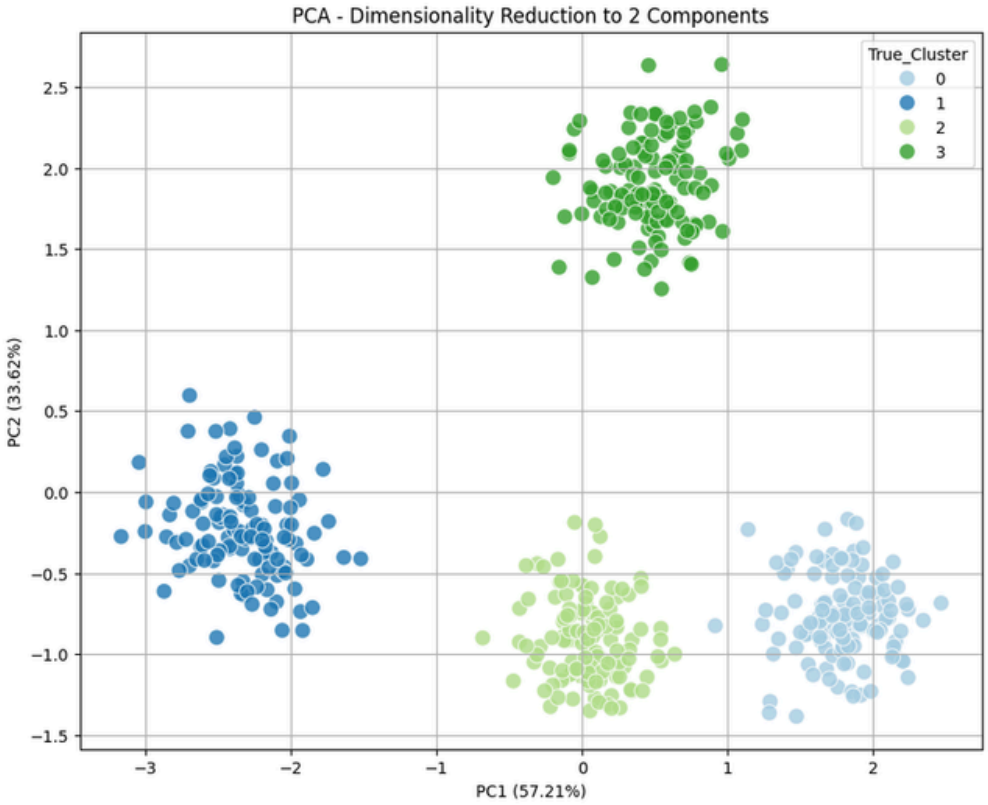
--- Part 2: Dimensionality Reduction with PCA ---

Original PCA Dataset Head:
   Feature_1  Feature_2  Feature_3  Feature_4  True_Cluster
0  -0.638667   1.110057  -6.400722  -0.204990             3
1  -2.951556  -7.657445   3.844794   0.903589             1
2  -0.253177   2.125103  -7.869801   0.559678             3
3  -2.151209   3.401400  -5.734930   0.965230             3
4  -2.347519  -7.230467   3.478891  -0.443440             1

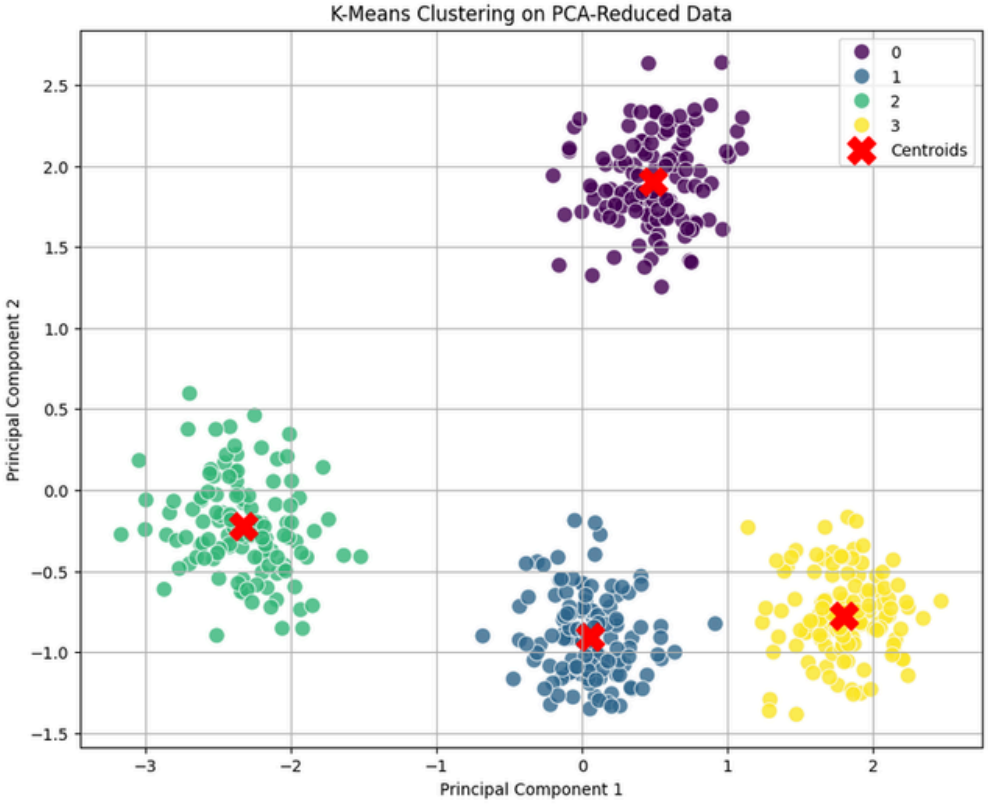
Original PCA Dataset Shape: (500, 5)

Principal Components Head:
   Principal_Component_1  Principal_Component_2  True_Cluster
0               0.455305               1.623917             3
1              -2.705622               0.375012             1
2               0.810234               1.966926             3
3               0.427139               2.149626             3
4              -2.407508               0.099250             1

Explained Variance Ratio: [0.57208431 0.33622342]
Total Explained Variance by 2 PCs: 0.908
```



--- Part 3: K-Means on PCA-Reduced Data ---



Silhouette Score for K-Means on PCA-Reduced Data (K=4): 0.776

Experiment completed successfully!